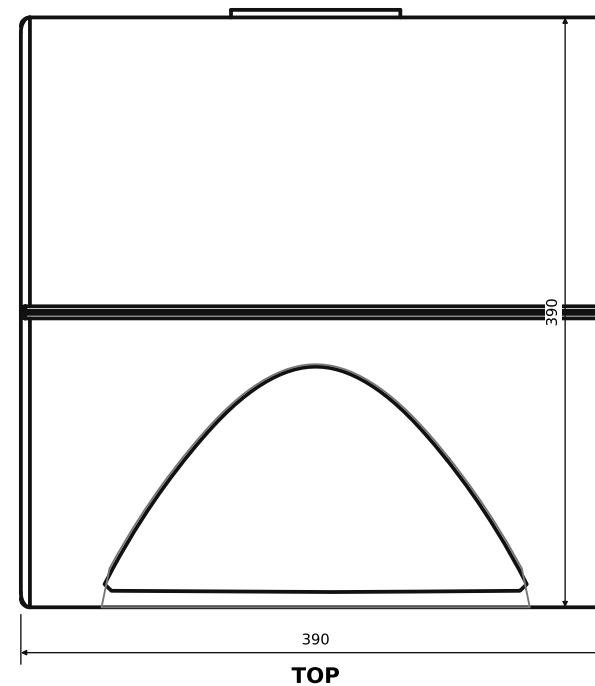
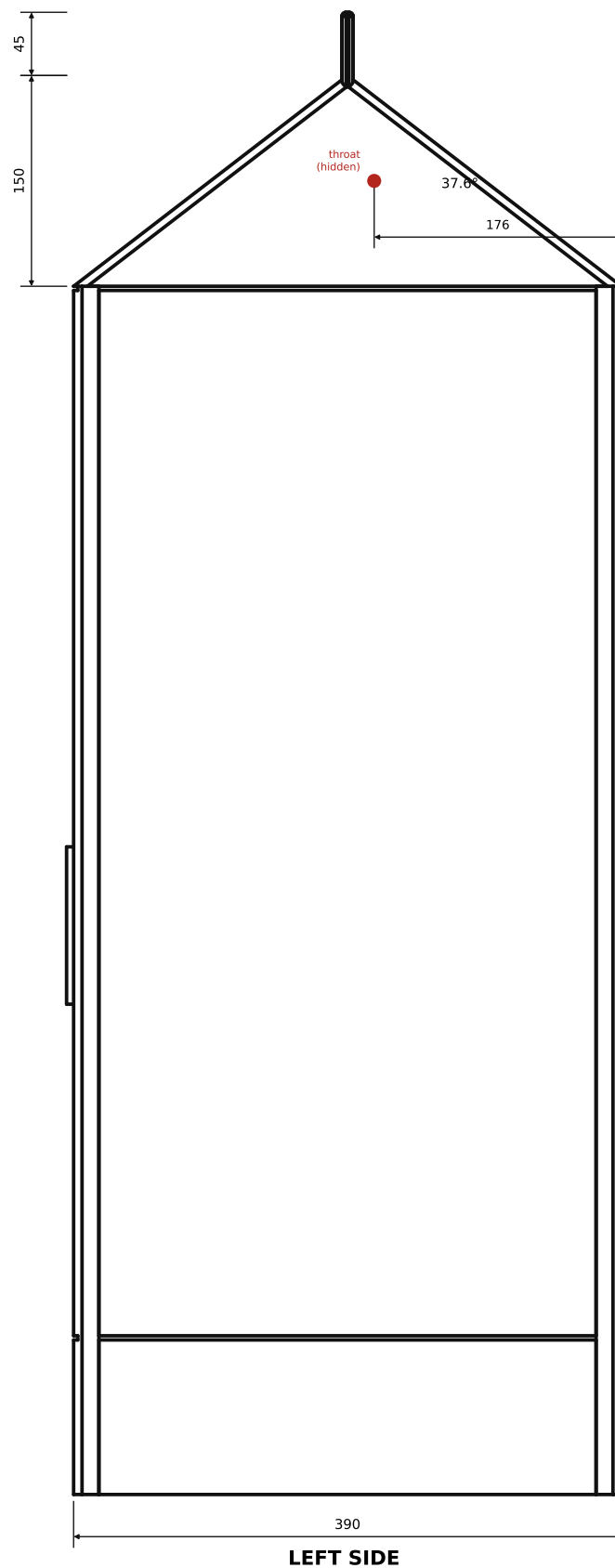
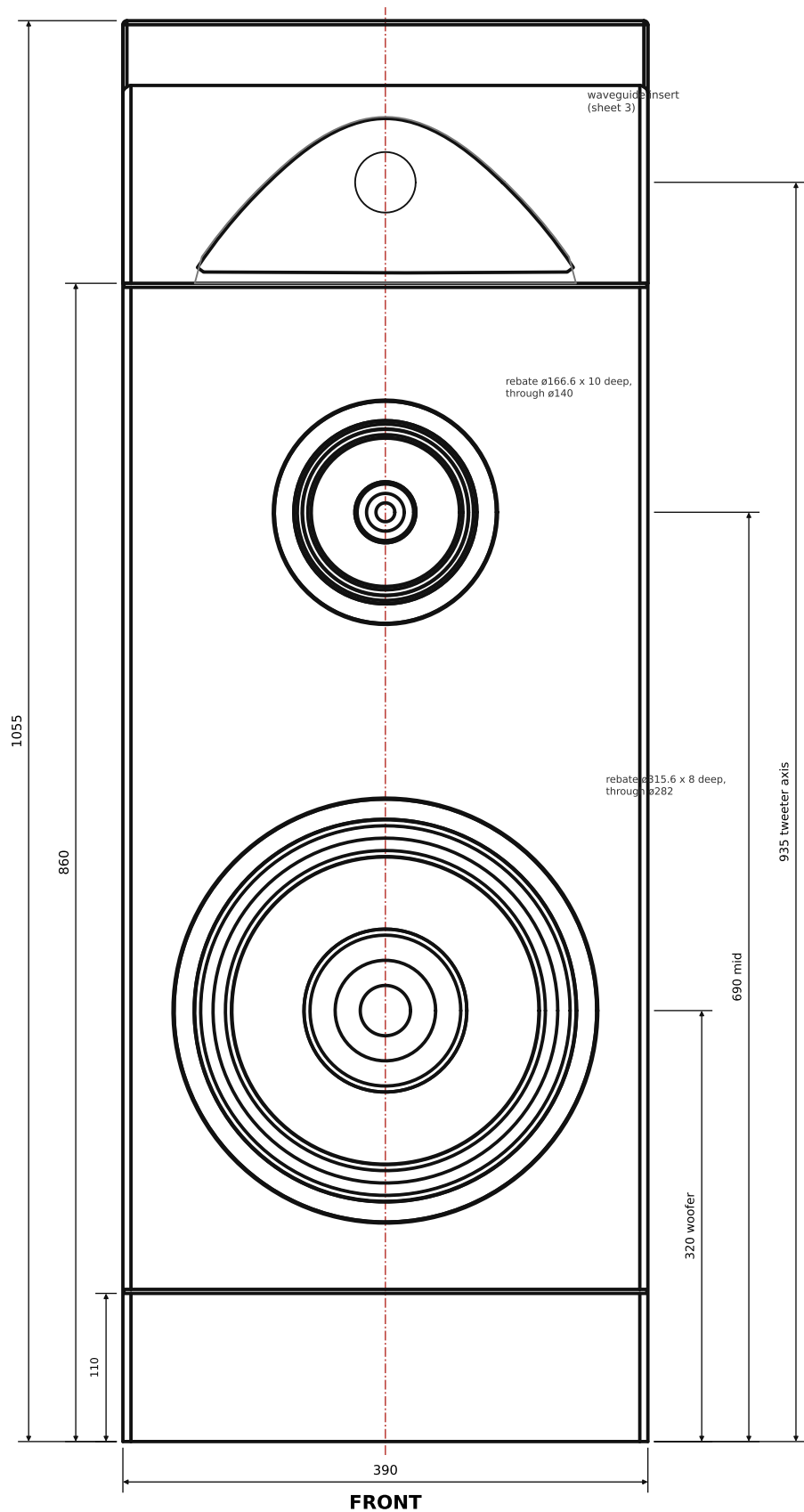




black: the waveguide's mouth, where its lip meets the roof
grey: the insert's seam

Notes

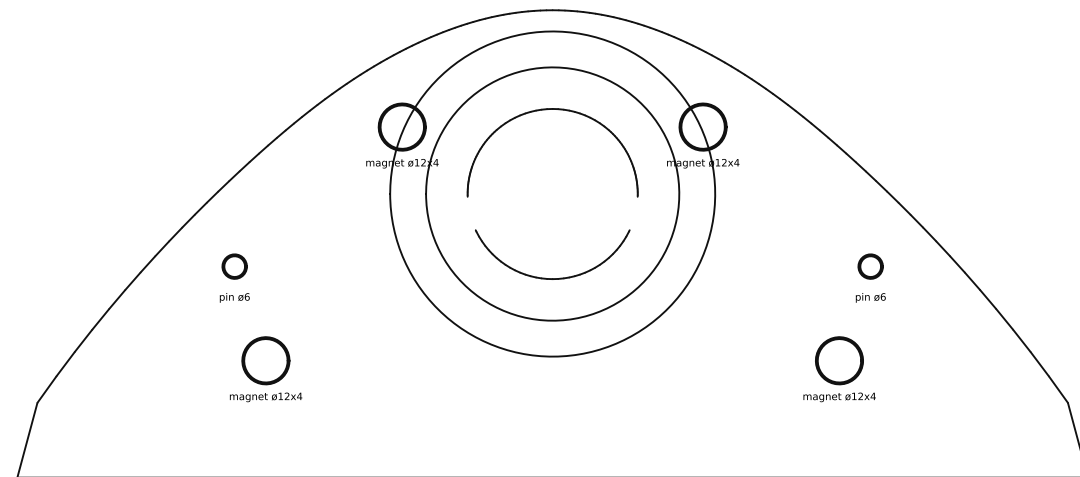
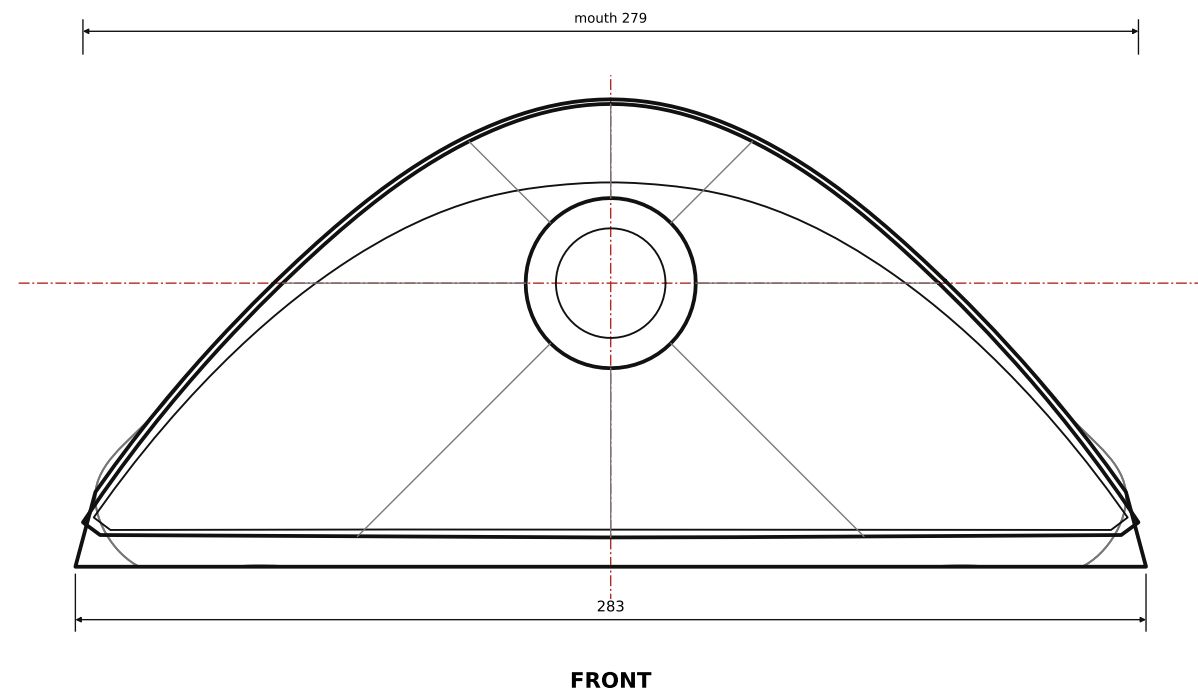
1. Cabinet: 18 mm Baltic birch, glued. The vertical corners and the gable's hips are rounded 6 mm, the fin's edges 3 mm. 3 x 3 mm shadow lines at z 110 and z 857.
2. The drivers sit flush: each frame in a rebate, a 3 mm trim ring over its flange and screws, level with the finish.
3. The tweeter is at the throat of a waveguide insert in the roof (sheet 3): axis z 935, 176 behind the front face. The shape is set by simulation (fab/out/acoustics/waveguide).
4. The back (port, amplifier, label) is drawn on sheet 4 with the wiring.
5. Finish: colour coat and 2K clear; the Facts printed between them.

earmilk floorstander | General arrangement

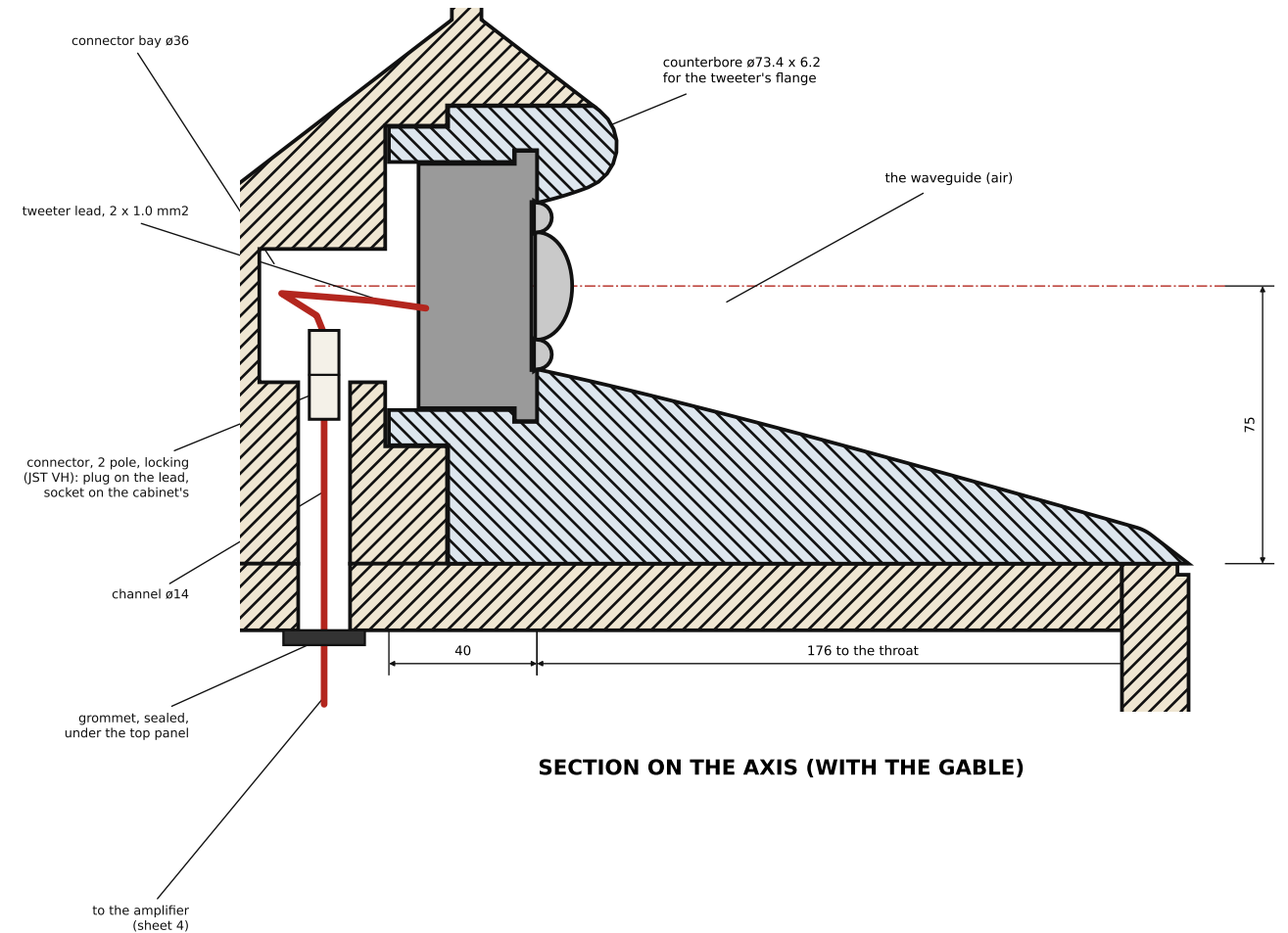
Front, left side and top at 1:5. Heights from the floor; depths from the front face.

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Check every bought part against its own drawing before cutting.



BACK (THE FACE THAT MEETS THE POCKET)



Fitting the tweeter and the insert

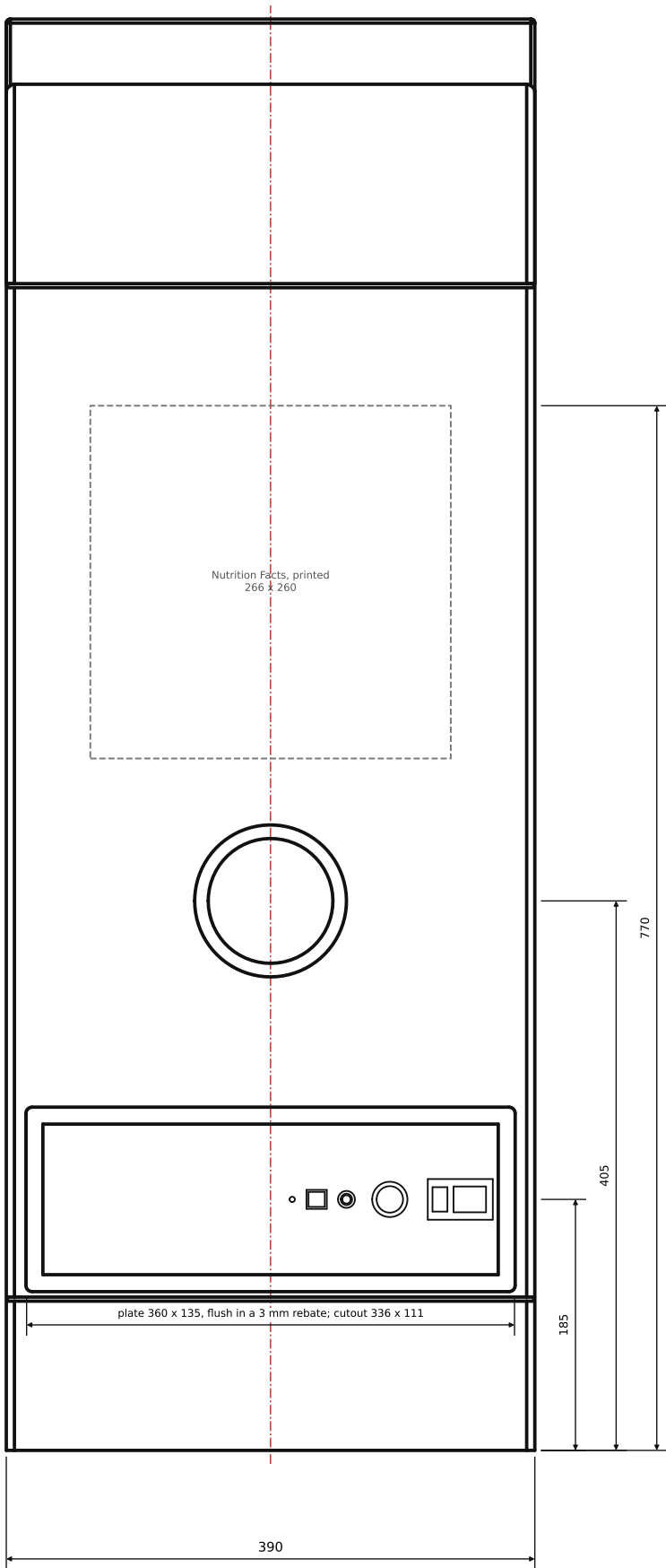
1. Solder the tweeter's lead (2 x 1.0 mm2, 120 mm) to its tabs and crimp the plug half of a two-pole locking connector (JST VH) on its end.
2. Seat the tweeter in the counterbore from behind, dome toward the throat, on a 0.5 mm foam gasket. Fix its 4 screws into the insert (heat-set brass inserts, M3).
3. Glue the magnets into the insert's back with epoxy, polarity marked, and their partners into the pocket's back wall, opposite poles out. Fit the two pins in the insert.
4. The cabinet's lead ends in the socket 100 mm above the grommet. With the insert out, draw the socket forward out of the bay through the empty pocket, plug the tweeter's lead into it, and tuck both and the slack back into the bay.
5. Slide the insert in, level, until the pins seat and the magnets pull it home: its face flush with the roof, the seam even.
6. Service: lift it out with a suction lifter on its flat border (or a pull loop under its lower edge), unplug, and the tweeter comes out with it.

earmilk floorstander | Waveguide insert and tweeter mount

The insert at 1:2: front, section on the axis, back; the pocket; fitting and service.

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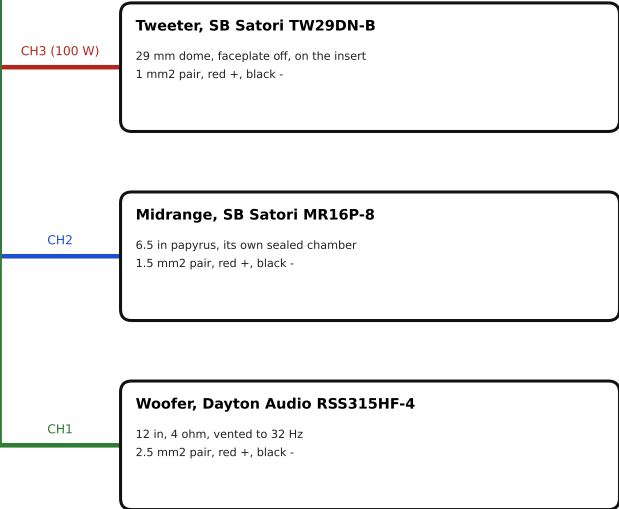


BACK

Mains
IEC C14 inlet, switch

Signal in
XLR (balanced) or RCA
USB for the DSP

Hypex FusionAmp FA253
3 channels, DSP crossover and EQ (ADAU1452)
250 + 250 W into 4 ohm, 100 W tweeter channel
on its side across the back's foot
in its own sealed box (not airtight itself)
filters: Hypex Filter Design, over USB
mains in on the plate (IEC C14 with switch)



Runs, connectors and seals

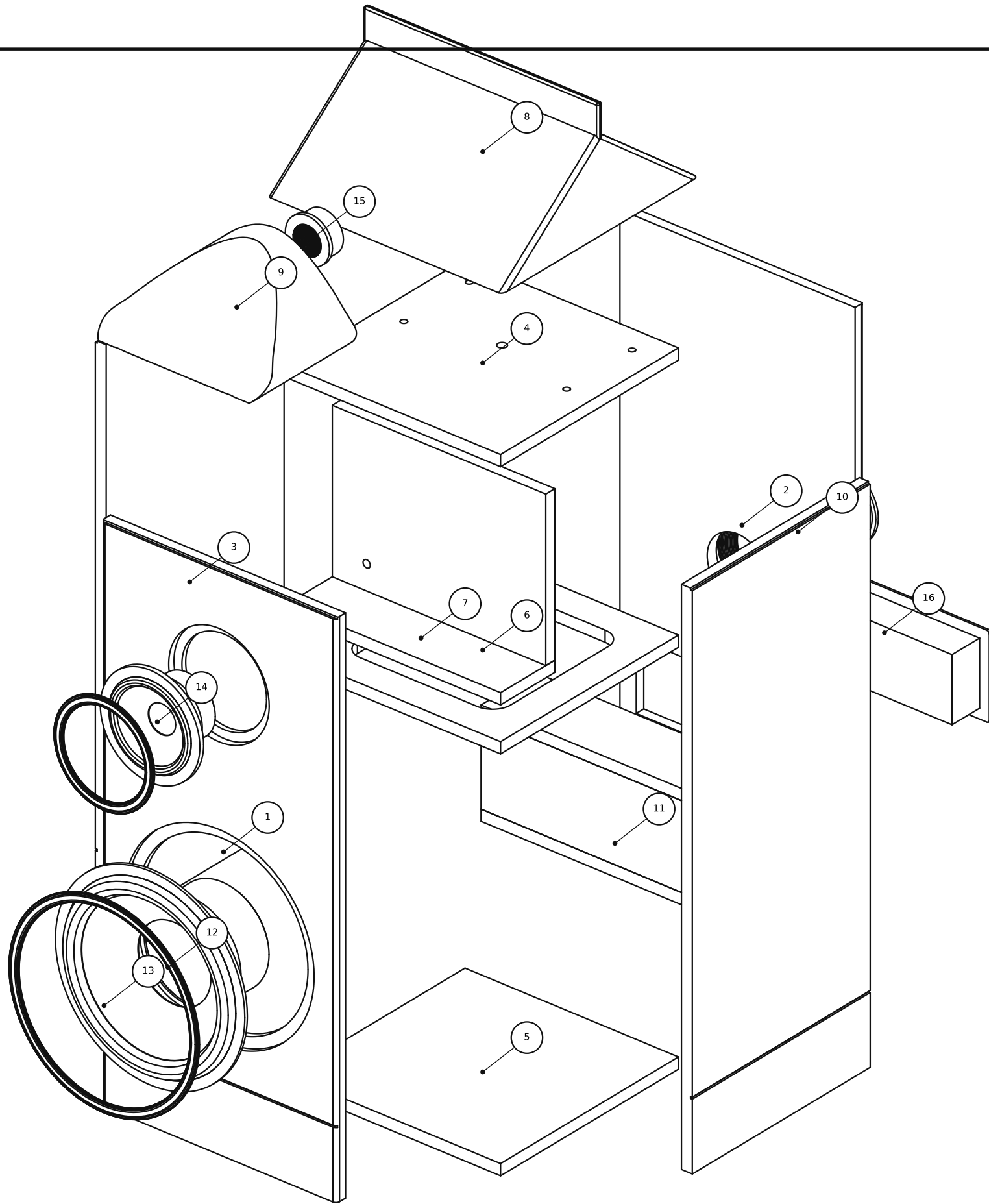
1. Woofer (CH1): from the amplifier box's gland straight to the woofer's terminals, 0.6 m of 2.5 mm2. Fit 6.3 mm push-on terminals to suit the driver.
2. Mid (CH2): up the back wall, through the 12 mm hole in the mid chamber's divider, 1.0 m of 1.5 mm2. Seal the hole round the cable with putty: the mid's chamber must stay closed.
3. Tweeter (CH3): from the gland up through the brace's window to the top panel, through the 14 mm channel in the top panel and the gable block, 1.3 m of 1.0 mm2, ending in the socket of a two-pole locking connector (JST VH) in the bay behind the insert. The supplied 1.25 m Hypex harness is too short: extend it.
4. Seal the tweeter cable where it passes the top panel with a rubber grommet and sealant. The insert's pocket is open to the room through its 0.3 mm seam, so an unsealed channel would be a leak in the woofer's box.
5. The amplifier box's lid: a 20 mm gland (PG11) for the three pairs; seal it. The box itself is glued and sealed; its front is the woofer chamber's wall.
6. Polarity, delay and the crossover (about 2.8 kHz tweeter to mid, from the waveguide study) are set in the DSP: wire red to + throughout and let the filters do the rest.

earmilk floorstander | Back and wiring

The back at 1:5 and the amplifier, drivers, cables and connectors. Active: the amplifier's DSP is the crossover.

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- Parts**
- 1 **Front baffle x1**
18 mm Baltic birch, CNC: cutouts and rebates for the drivers
 - 2 **Back x1**
18 mm birch: port hole, the amplifier's cutout and rebate
 - 3 **Sides x2**
18 mm birch
 - 4 **Top panel x1**
18 mm birch, the tweeter cable's 14 mm hole, four dowel holes
 - 5 **Bottom panel x1**
18 mm birch
 - 6 **Window brace x1**
18 mm birch
 - 7 **Mid chamber shelf and divider x2**
18 mm birch; the divider's 12 mm cable hole
 - 8 **Gable block x1**
laminated birch, CNC: the insert's pocket and the cable channel
 - 9 **Waveguide insert x1**
CNC from solid (MDF, HDPE) or printed (SLA, MJF); magnets, pins
 - 10 **Port x1**
printed tube and flare collar, 92 bore
 - 11 **Amplifier box x3**
18 mm birch floor, lid (gland) and front, glued and sealed
 - 12 **Woofer x1**
Dayton Audio RS5315HF-4, 12 in, 4 ohm, black anodised aluminium cone
 - 13 **Trim rings x2**
printed, satin black or gunmetal, over each frame and its screws
 - 14 **Midrange x1**
SB Acoustics Satori MR16P-8, 6.5 in papyrus cone
 - 15 **Tweeter x1**
SB Acoustics Satori TW29DN-B, faceplate off, rear-mounted on the insert
 - 16 **Amplifier x1**
Hypex FusionAmp FA253, DSP, 3 channels

earmilk floorstander | Exploded view and parts

The parts pulled apart along the way they go together, and what each one is.

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